

UPSHUR COUNTY REGIONAL, WV, USA

WMO: 720328

Lat: 39.000N

Lon: 80.274W

Elev: 498

StdP: 95.48

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63832

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-12.1	-19.9	0.7	-12.7	-17.3	0.9	-9.9	10.8	6.0	9.3	6.8	2.3	280	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	31.2	22.7	29.9	22.2	28.7	21.5	24.2	28.4	23.6	27.4	22.9	26.4	2.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.8	25.4	22.5	18.3	25.1	22.2	17.9	24.9	76.0	28.5	72.9	27.8	71.0	26.6	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	8.8	7.6	6.7	DB	-17.6	32.8	3.1	1.0	-19.8	33.5	-21.6	34.0	-23.4	34.6	-25.7	35.3
(5)				WB	-18.1	25.6	3.0	1.3	-20.3	26.6	-22.0	27.4	-23.7	28.1	-25.9	29.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	0.3	1.2	6.1	12.2	16.9	20.6	22.5	21.9	19.3	12.7	6.9	3.3
	DBStd	9.41	7.25	6.87	6.37	5.49	4.54	2.99	2.43	2.59	3.80	5.17	5.44	5.86
	HDD10.0	1140	310	256	157	39	4	0	0	0	0	31	123	221
	HDD18.3	2743	559	479	379	192	82	14	2	4	34	186	344	467
	CDD10.0	1888	9	9	36	107	219	317	387	368	278	115	29	13
	CDD18.3	448	0	0	1	9	39	81	131	113	62	11	1	0
	CDH23.3	3311	0	1	8	98	324	574	920	794	496	92	3	0
	CDH26.7	832	0	0	0	16	61	134	256	210	137	18	0	0
Wind	WSAvg	2.2	3.0	3.1	2.9	2.9	2.0	1.8	1.5	1.3	1.3	1.9	2.3	2.7
Precipitation	PrecAvg	1190	85	74	109	100	117	113	123	102	98	81	93	91
	PrecMax	1645	151	173	235	176	326	256	236	208	208	206	294	204
	PrecMin	860	28	24	36	27	36	31	44	44	9	9	34	23
	PrecStd	168	36	31	47	37	60	52	45	37	41	44	46	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.8	21.3	24.0	28.7	30.1	31.4	32.5	32.3	32.2	28.8	22.8	21.0
		MCWB	13.6	15.4	14.7	15.4	20.7	21.9	23.8	23.4	21.4	21.1	15.0	16.1
	2%	DB	16.3	17.4	21.4	26.2	27.9	29.8	31.1	30.8	29.9	26.0	20.1	17.4
		MCWB	12.1	12.2	13.5	15.4	19.5	22.1	23.7	23.0	21.5	18.3	13.5	12.8
	5%	DB	13.8	14.1	18.7	23.8	27.2	27.9	29.0	28.9	27.9	23.1	17.7	14.9
		MCWB	10.7	10.0	12.3	14.2	19.2	21.3	22.6	22.1	20.5	16.9	12.0	11.7
	10%	DB	12.1	12.4	16.3	22.0	25.1	27.1	27.8	27.6	26.4	21.2	16.0	12.7
		MCWB	10.0	9.4	10.7	13.2	18.0	20.8	22.0	21.5	19.4	16.7	11.0	10.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	15.9	16.2	18.3	22.9	24.2	25.6	25.0	24.0	22.4	17.1	16.2
		MCDB	17.4	19.0	21.8	24.8	27.2	28.2	29.8	29.3	28.5	25.5	21.6	19.3
	2%	WB	12.9	13.8	14.8	17.1	21.3	23.3	24.5	24.1	22.9	20.5	15.1	14.0
		MCDB	15.3	16.5	19.3	22.3	26.0	27.2	28.7	28.0	26.9	23.7	18.3	16.1
	5%	WB	11.6	11.9	13.6	15.9	20.4	22.5	23.7	23.3	21.9	18.8	13.3	12.4
		MCDB	13.5	13.9	17.2	20.8	24.9	26.4	27.6	26.9	25.3	22.4	16.2	14.3
	10%	WB	9.4	8.9	12.1	14.8	19.4	21.7	23.0	22.5	20.9	17.1	11.8	10.6
		MCDB	12.1	11.8	15.5	19.6	23.5	25.5	26.2	25.9	24.6	19.7	14.9	12.6
Mean Daily Temperature Range	5% DB	MDBR	8.4	8.9	10.0	11.9	11.0	10.3	9.9	10.3	11.2	10.5	9.7	8.3
		MCDBR	10.7	12.4	13.3	14.7	13.3	11.8	11.6	11.9	12.9	13.5	12.9	11.9
		MCWBR	7.9	9.3	7.9	7.0	6.2	5.4	4.9	4.8	5.4	6.8	8.1	8.7
	5% WB	MCDBR	9.8	11.5	11.6	11.6	10.9	9.7	9.9	9.6	10.5	11.3	10.6	11.1
		MCWBR	7.8	10.0	7.8	6.6	5.5	5.0	4.9	4.5	5.0	6.1	7.9	9.0
Clear-Sky Solar Irradiance	taub	0.296	0.314	0.339	0.376	0.419	0.446	0.467	0.482	0.390	0.352	0.316	0.293	
	taud	2.482	2.418	2.418	2.343	2.225	2.185	2.148	2.074	2.328	2.435	2.510	2.525	
	Ebn at Noon	885	910	919	900	863	837	815	791	853	854	853	863	
	Edh at Noon	78	95	106	122	140	146	150	158	115	93	76	70	
All-Sky Solar Radiation	RadAvg	1.71	2.49	3.56	4.70	5.29	5.86	5.62	5.21	4.35	3.10	2.16	1.51	
	RadStd	0.20	0.37	0.27	0.39	0.45	0.40	0.33	0.28	0.48	0.38	0.26	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+59

Nomenclature: See separate page